

A. **Euclidean Algorithm.** There is an algorithm that outputs the gcd.

At its theoretical core is the following result.

Theorem. Let a and b be integers, with b nonzero.

If r is the remainder upon division $b \div a$ then:

$$\gcd(a, b) =$$

$$\text{i.e. } \gcd(\text{divisor}, \text{dividend}) =$$

Proof. We apply the division algorithm to write:

$$b = qa + r$$

Our goal is to show:

We do this by showing that the common divisors of a and b are the same as:

in which case, their greatest common divisors will be the same.

Every common divisor of a and r also divides b because:

Thus, every common divisor of a and r is also a common divisor of a and b .

Every common divisor of a and b also divides r because:

Thus, every common divisor of a and b is also a common divisor of a and r .

□

Euclidean Algorithm. Let $a < b$ be positive integers.

Initialization. Pick: $[\text{dividend}] =$

Pick: $[\text{divisor}] =$

Find the remainder upon division $[\text{dividend}] \div [\text{divisor}]$.

Stopping Condition. Observe:

$$\gcd(a, b) = \gcd(\text{divisor}, \text{dividend}) =$$

and thus, if the remainder is zero:

$$\gcd(a, b) =$$

If the remainder is not zero, then enter the cycle.

Cycle. Pick: $[\text{new dividend}] =$

Pick: $[\text{new divisor}] =$

Find the remainder upon division $[\text{new dividend}] \div [\text{new divisor}]$.

Assess the stopping condition, and re-cycle if it is not met.

Example 1. Use the Euclidean algorithm to calculate $\gcd(7544, 230)$.

Solution. We begin the algorithm.

Initialization. Execute:

The remainder is not zero.

So the divisor becomes our new dividend, and the remainder our new divisor.

We execute:

And we continue to cycle.

We execute:

Now the remainder is zero. Thus the gcd is our final divisor.

$\gcd(7544, 230) =$

B. **Solving Bézout.** The algorithm can be reversed to solve Bézout's identity.

Working backwards, the Euclidean algorithm yields integers x and y so:

$$\gcd(a, b) =$$

We demonstrate this using our calculation of $\gcd(7544, 230) = 46$.

$$7544 = 32 \cdot 230 + 184$$

$$230 = 1 \cdot 184 + 46$$

$$184 = 4 \cdot 46 + 0$$

Start from the penultimate step where the gcd is the remainder.

Solve for the gcd in terms of the dividend and divisor.

Go up one line to solve for the prior remainder in terms of its dividend/divisor.

Substitute this into your expression for the gcd.

We've done it.

Example 2. Use the Euclidean algorithm to find integers x, y, z so:

$$\gcd(7544, 230, 96) = 7544x + 230y + 96z$$

Solution. First an aside: we discuss the gcd of three nonzero integers a, b, c .

Observe that every common divisor of a, b, c is a common divisor of:

and vice versa.

Thus:

Now, we return to the main problem, concluding:

$$\gcd(7544, 230, 96) =$$

Next we apply Euclidean algorithm to calculate this gcd.

$$\text{So: } \gcd(7544, 230, 96) =$$

Next we reverse the algorithm.

Finally we use the result from the last page:

$$46 = -1 \cdot 7544 + 33 \cdot 230$$

to finish.